AI Ethics & Cybersecurity — One-Page Cheat Sheet

Unit 2 (Fairness, Bias & GenAI) + Module 3 (Ethics & Cybersecurity Principles) · purple = keyword to write in answers

★★★ Must-know ★★ Important ★ Quick read

0. Basics: Bias vs Fairness U2 ★★

Bias = systematic error → unfair decisions (a.k.a. ML/algorithmic bias); sources: data collection, algorithm design, human bias. **Fairness** = correcting algorithmic bias so outcomes are just & equitable w.r.t. **sensitive/protected attributes** (race, gender, religion, age, SES).

Types of bias (8): Sampling · Algorithmic · Confirmation · Measurement · Generative · Reporting · Automation (trust machine over human) · Group-attribution (*in-group*, *out-group homogeneity*). Pipeline view: data (historical, representation, measurement) → modeling (evaluation, aggregation) → human review.

Types of fairness (5): **Group** (groups treated equally) · **Individual** (similar people → similar outcome) · **Procedural** (fair, transparent process) · **Counterfactual** (same result if attribute flipped) · **Causal** (no decisions from historical inequality).

Principles: Non-discrimination, Equitable treatment, Transparency, Accountability, Ethical considerations.

Impact — fair AI: trust, equal opportunity, legal compliance, social cohesion, business benefit. **Biased AI:** reinforces inequity, loss of trust, discrimination (loans/jobs), legal risk, economic & social cost.

	Bias	Fairness
Nature	often unintentional, technical	deliberate, intentional goal
Focus	accuracy/reliability	equitable treatment
Measured by	accuracy, precision, recall	demographic parity, equal opp.

1. Techniques for Measuring Fairness & Bias U2 ★★★

1. **Disparate Impact Analysis** – does AI hurt one group more? Procedure: group fairness, statistical tests, thresholds. Used in hiring, lending, insurance.
2. **Fairness Through Awareness** – fairness metrics built into training/eval: Demographic Parity, Equal Opportunity, Equalized Odds.
3. **Counterfactual Fairness** – same outcome if sensitive attribute changed (individual fairness, counterfactuals).
4. **Explainability & Interpretability** – feature importance, local explanations (LIME/SHAP).
5. **Bias Detection Tools** – bias metrics + auditing tools: **AI Fairness 360** (IBM), **What-If Tool** (Google).
6. **Human-in-the-Loop** – expert judgment, feedback loops.
7. **Ethical Impact Assessment** – stakeholder consultation, risk assessment, mitigation strategies.

Metric	Condition (equal across groups)
Demographic / Statistical Parity	P(Ŷ=1 | A=a) = P(Ŷ=1 | A=b) — same positive rate
Equal Opportunity	same **TPR** (true-positive rate)
Equalized Odds	same **TPR and FPR**
Disparate Impact (DI)	DI = P(Ŷ=1|unpriv) ÷ P(Ŷ=1|priv); **DI < 0.8 → biased** (4/5ths / 80% rule)

Ex: loan approval women 30%, men 50% → DI = 0.6 < 0.8 ⇒ disparate impact.

🔑 disparate impact · 80% rule · TPR/FPR · counterfactual · AIF360 · What-If Tool · human-in-the-loop

2. Techniques for Mitigating Bias U2 ★★★

Pre-processing → In-processing → Interpretability → Post-processing

1. **Data collection & pre-processing:** diverse & representative data; bias detection (statistical parity, disparate impact); augmentation, anonymization, careful sampling, **reweighing**.
2. **Algorithmic fairness (in-processing):** **fairness constraints** on sensitive attributes; **fairness-aware algorithms** optimizing equal opportunity / equalized odds; regularization.
3. **Interpretability & transparency:** **XAI**; regular **bias audits**.
4. **Post-processing:** reweighing, demographic parity adjustment, **disparate impact remover**, threshold adjustment — applied after training.

7-step holistic approach: Data & preprocessing → Algorithm design → Fairness-aware training → Evaluation/testing (bias audits) → Transparency/XAI → Monitoring & accountability → Policy, regulation & governance.

3. Policies & Frameworks for Fairness U2 ★★

Policies

- Ethical AI principles: fairness, accountability, transparency
- **GDPR**: rules on automated decision-making (Art. 22)
- **Algorithmic Accountability Act** (US, *proposed*)
- **IEEE P7003** – algorithmic bias considerations
- Industry-association AI ethics guidelines

Frameworks

- Definitions: group, individual, **intersectional** (multiple attributes together)
- Metrics: disparate impact, equal-opportunity difference; evaluation protocols over lifecycle
- Continuous monitoring: feedback loops, stakeholder engagement

4. Bias in Data Collection U2 ★★

Systematic errors introduced when data is gathered/processed → unfair outcomes.

Type	Meaning	Example
Selection	groups excluded / under-represented	recruitment data missing some demographics
Sampling	selection method favours outcomes	poll over-represents one party
Measurement	tools mis-measure some groups	scales calibrated differently
Reporting	selective reporting	negative drug trials unpublished

Causes: historical bias · algorithmic bias (facial recognition less accurate on some groups) · cultural/contextual bias. **Impact:** unfair treatment, inaccurate predictions, loss of trust. **Mitigate:** diverse data, bias detection & correction, transparency/documentation, community engagement.

5. Fairness in Data Processing U2 ★★

Techniques: bias detection & mitigation (demographic parity, equalized odds, DI analysis) · algorithmic fairness · model interpretability (XAI, auditable) · **data governance** & ethics frameworks.

Challenges: fairness definitions complex & context-dependent · data-quality issues · **fairness vs utility trade-off**.

Best practices: diverse stakeholder engagement · continuous monitoring & evaluation · education & training.

6. Generative AI U2 ★

AI that **learns patterns** from data and **generates new content** (text, image, video, music, 3D). Driven by **LLMs** / **foundation models**.

Works in 3 phases: ① **Training** → foundation model on huge data ② **Tuning** → **fine-tuning** (labeled task data) / **RLHF** (humans score outputs) ③ **Generation, evaluation, re-tuning** (+ **RAG** = retrieve external docs → generate; current & verifiable).

Architectures: VAE (compress features → variations; 2013) · **GAN** (generator vs discriminator; 2014) · **Diffusion** (remove noise step-by-step; DALL·E) · **Transformer** (**attention**; "Attention Is All You Need" 2017; basis of GPT).

Key concepts: generative models · **latent space** (hidden representation; sample → diverse outputs) · **backpropagation** (find where error came from) · **gradient descent** (reduce error).

Timeline: ELIZA 1964 → GPU (GeForce 256) 1999 → Autocomplete 2004 (Markov chain) → VAE 2013 → GAN 2014 → Transformer 2017 → GPT-2/3 2019-20 → ChatGPT 2022 → Gemini, Copilot, Llama, watsonx.

Applications: creative arts, content/marketing, healthcare (synthetic images), gaming/VR, personalization. **Advantages:** creativity, faster decisions, dynamic personalization, 24×7.

Risks: **hallucination** (fix: guardrails, RAG) · inconsistent output (prompt engineering) · bias · black box (XAI) · security/privacy/IP · **deepfakes**. **GenAI vs AI:** creates new content vs predicts/classifies.

7. Types of Bias in Generative AI U2 ★★★

Unintended, systematic distortions in generated content reflecting bias in training data or model.

Type	Example → rule
Dataset	"engineer" images mostly men → *biased data → biased output*
Sampling	mostly young people's opinions → *less diverse output*
Representation	a gender always shown in one job → *stereotypes*
Cultural	Western-trained AI misreads Indian practices → *limited perspective*
Algorithmic	design over-weights certain features → unfair output
Stereotype amplification	repeats & strengthens existing stereotypes

Prevention (7): ① diverse data (collection, sampling, augmentation) ② bias detection (pre-training analysis, reweighting) ③ fairness-aware algorithms (metrics in objective/regularization) ④ transparency & XAI + documentation ⑤ ethical guidelines & governance, diverse stakeholders ⑥ continuous post-deployment monitoring, feedback loops ⑦ education & awareness. (+ human-in-the-loop review)

Mnemonic **"DS-RCAS"**: Dataset, Sampling, Representation, Cultural, Algorithmic, Stereotype.

8. Importance of Privacy & Security in AI M3 ★★

- **Protect personal info** → prevents identity theft, fraud
- **Build trust** → users adopt trusted AI
- **Regulatory compliance** → GDPR/HIPAA; avoid fines
- **Ethical responsibility** → no misuse of data
- **Prevent discrimination** → anonymize, reduce bias
- **Model integrity** → defend against AI threats: **data poisoning**, **adversarial examples**, **model inversion**, **membership inference**, model theft, **prompt injection**

AI + Data ⇒ Privacy (protect) + Ethics (use responsibly) + Fairness (avoid unfair outcomes)

CIA triad: Confidentiality · Integrity · Availability. **Privacy by design**, data minimisation, encryption, access control.

9. AI-Specific Security Tools & Software M3 ★★

Category	Purpose	Tools
IDPS	detect/prevent suspicious network activity	Darktrace, Vectra
Endpoint security	protect laptops, servers, phones	Cylance, CrowdStrike Falcon
SIEM	collect & analyse logs/events	Splunk, IBM QRadar
UEBA	unusual user behaviour, insider threats	Exabeam, Securonix
Threat intel	info on cyber threats	Recorded Future, ThreatConnect
Incident response (SOAR)	automate/orchestrate response	Cortex XSOAR, Swimlane
Fraud detection	suspicious transactions	Kount, Sift Science
Phishing detection	block malicious emails	Area 1, Cofense
IAM	who accesses what; adaptive MFA	SailPoint, Okta

Mnemonic **"I-E-S-U-T-I-F-P-I"** → "**I** **E**at **S**ome **U**nusual **T**acos, **I** **F**eel **P**retty **I**ll"

10. PPML – Privacy-Preserving ML M3 ★★★

Train useful models **without exposing individual data**; balance **utility vs privacy**.

Technique	Idea	Example
Federated Learning	train on devices; raw data stays local; only **model updates** sent; server aggregates (FedAvg)	3 hospitals train locally; Gboard keyboard
Differential Privacy	add calibrated **noise** so one person's presence/absence doesn't change result; privacy budget **ε** (smaller ε = more privacy)	avg salary 50,000 → released 50,120
Homomorphic Encryption	compute on **encrypted** data without decrypting (partial / fully FHE)	cloud analyses encrypted patient records
SMPC (Secure Multi-Party Computation)	parties jointly compute a function without revealing inputs (secret sharing)	hospitals total 100+150+200=450, counts hidden

Others: **secure enclaves / TEE**, synthetic data, data anonymization before training.

11. PPDM – Privacy-Preserving Data Mining M3 ★★★

Extract useful **patterns** while hiding sensitive info.

- **Anonymization** (generalization/suppression; *Age 23 → 20-25, Noida → NCR, names removed*):
  - **k-anonymity**: each record indistinguishable from ≥ k−1 others ("who is this?")
  - **l-diversity**: ≥ l distinct sensitive values per group (fixes homogeneity attack)
  - **t-closeness**: group's sensitive distribution within t of overall (fixes skew attack)
- **Data perturbation / randomization**: add noise/swap values (20,21,22,23 → 20,22,21,24), patterns kept
- **Cryptographic**: encryption, HE, SMPC for distributed (horizontally/vertically partitioned) data
- **Rule hiding**: suppress sensitive association rules

PPML	PPDM
protect privacy while **learning/training models** (FL, DP, HE, SMPC)	protect privacy while **mining patterns** (anonymize, perturb, crypto)

Strength: k < l < t. Trade-off: **more privacy = less data utility**.

12. Risk Management: Assessment & Incident Response M3 ★★★

Risk Assessment – identify, evaluate, prioritize risks

1. **Asset & threat identification**: hackers, malware, insiders, disasters, AI attacks
2. **Vulnerability assessment**: bugs, misconfigs, human error, weak models
3. **Impact analysis**: operations, reputation, finance, assets
4. **Risk evaluation**: **Risk = Likelihood × Impact (Severity)** → risk matrix (High/Med/Low)
5. **Mitigation / treatment**: technical controls, policy, training. Options: **Avoid · Reduce · Transfer (insurance) · Accept**
6. **Monitor & review** continuously

Incident Response (NIST SP 800-61 lifecycle)

① **Preparation** → ② **Detection & Analysis** → ③ **Containment, Eradication & Recovery** → ④ **Post-Incident Activity (lessons learned)** ↻

Preparation = IR plan, team (CSIRT), drills/tools. Detection = SIEM/IDS alerts, triage. Containment = isolate systems; Eradication = remove malware/root cause; Recovery = restore from backup, verify. Post-incident = review, report, update plan.

Goal: minimize damage, restore operations quickly, prevent recurrence. AI frameworks: **NIST AI RMF** (Govern, Map, Measure, Manage).

13. Regulatory Compliance: GDPR vs HIPAA M3 ★★★

	GDPR (General Data Protection Regulation)	HIPAA (Health Insurance Portability & Accountability Act)
Region / year	EU/EEA · effective **25 May 2018**	USA · **1996**
Protects	any **personal data** of individuals	**PHI** – Protected Health Information (e-PHI)
Applies to	any org processing EU residents' data, **anywhere** (extraterritorial)	**Covered entities** (providers, health plans, clearinghouses) + **business associates** (BAA)
Key rules	lawful basis/**consent**; principles: lawfulness, purpose limitation, data minimisation, accuracy, storage limitation, integrity & confidentiality, accountability	**Privacy Rule** (use/disclosure, minimum necessary) · **Security Rule** (administrative, physical, technical safeguards) · **Breach Notification Rule**
Rights	access, rectification, **erasure (right to be forgotten)**, portability, object, **no solely automated decisions (Art. 22)** → right to explanation	patients can access/obtain copies & request corrections of their records
Breach notice	to authority within **72 hours**	to individuals within **60 days**
Penalty	up to **€20 M or 4%** of global turnover	tiered civil & criminal penalties
Tools	**DPIA**, DPO, privacy by design	de-identification (Safe Harbor: remove **18 identifiers**)

AI relevance: lawful training data, consent, explainability, anonymize/pseudonymize, audit logs, encryption.

14. Ethics Across the AI Lifecycle M3 ★★

Problem definition (stakeholders, purpose, impact assessment) → **Data collection** (privacy GDPR/CCPA, informed consent, diverse data) → **Model development** (transparency, bias mitigation, accountability) → **Evaluation** (fairness across groups, adversarial robustness, explainability) → **Deployment** (monitoring, user training, feedback) → **Operation & maintenance** (ongoing evaluation, model updates, ethical audits).

15. Case Studies: Implementing AI Ethics in Engineering M3 ★★

Google AI Principles (2018): socially beneficial, avoid unfair bias, safe, accountable, privacy, scientific excellence. *Implementation:* bias audits, diverse data, safety testing, AI ethics review board. *Outcome:* more trust, transparency, accountability, less bias.

IBM AI Fairness 360: open-source toolkit, 70+ metrics & mitigation algorithms (reweighing, adversarial debiasing, reject-option). **Microsoft Responsible AI**: fairness, reliability & safety, privacy & security, inclusiveness, transparency, accountability.

Failures (lessons): **Amazon hiring tool** (penalised women – trained on male-dominated CVs; scrapped) · **COMPAS** (recidivism scores biased vs Black defendants) · facial recognition higher error on darker-skinned women.

Other frameworks: PAVED (Partnership on AI – transparency, fairness, accountability) · **IEEE Global Initiative – Ethically Aligned Design** · EU AI Act (risk-based).

Best-practice checklist: ethics review → DPIA/impact assessment → diverse data & documentation (datasheets, model cards) → fairness testing → XAI → security testing (red-team) → human oversight → monitoring & audits.

LAST-MINUTE STRIP: Metrics: DP · EO(TPR) · EOdds(TPR+FPR) · DI<0.8 × | Mitigate: Pre → In → Post | GenAI bias: Dataset · Sampling · Representation · Cultural · Algorithmic · Stereotype | PPML: FL (data local) · DP (noise, ε) · HE (compute encrypted) · SMPC (compute jointly) | PPDM: k / l / t + perturb + crypto | Risk = L × I; Avoid/Reduce/Transfer/Accept | IR: Prepare → Detect → Contain/Eradicate/Recover → Lessons | GDPR = EU personal data, 72 h, €20M/4% | HIPAA = US PHI, Privacy/Security/Breach rules, 60 days